Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/07/2011
PI Name:	Dynlacht, Joseph Robert Ph.D.		
Project Title:	Effects of Estrogen on Cataract Induction After Exposure to High LET Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202-5121	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	09/01/2005	**End Date:**	05/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 5/31/2011 per C. Guidry/JSC (10/2010) NOTE: PI to receive NCE to 8/31/2010 per J. Dardano/JSC (7/09)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Mendonca, Marc (Indiana U) Valluri, Shelia (Indiana U) Bigsby, Robert (Indiana U)		
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Performance Goal No.:			
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Task Description:	Several studies have demonstrated that there is an increased cataratogenic potential in animals and humans exposed to high linear energy transfer (LET) radiation compared to low LET radiation. Since the relative biological effectiveness (RBE) of heavy energetic charged (HZE) particles for lens opacification is high, these studies imply a high cataract risk for astronauts involved in long-duration space missions. During a manned mission to Mars, ocular doses in excess of ~0.1 Gy of HZE particles could be accumulated. Though not directly life-threatening, radiation-induced cataractogenesis represents a potentially serious sequelae which could necessitate surgical intervention in the years after, or perhaps even during the mission in some individuals (see below). While the cellular and molecular mechanism(s) of radiation-induced cataractogenesis have not been clearly elucidated, damage to DNA at the time of exposure and subsequent proliferation of the radiosensitive cells in the germinative zone of the lens epithelium likely play a role in the process. Using a rat model, we have recently accumulated data which indicate that estrogen increases or reduces the latent period and may modulate the incidence and severity of cataracts induced by low LET radiation, depending upon when the estrogen is administered. High estrogen levels are induced in non-pregnant women using oral contraceptives, or in post-menopausal women on estrogen replacement therapy. Therefore, sub-groups of female astronauts may be at an increased risk for developing cataracts that are more severe or occur with a more rapid onset. The interaction between estrogen and high LET radiation has never been studied before in the context of cataractogenesis. Therefore, a mechanistic approach characterizing the modulating effect of estrogen on high LET radiation-induced cataractogenesis is proposed. The underlying hypothesis of this proposal is that estrogen alters the latent period for high LET radiation-induced cataractogenesis, and, depending upon when the it is administered, modulates severity and incidence of cataracts. Estrogens regulate several proteins involved in cell cycle control and apoptosis, and it's metabolism results in the production of free radicals which may be genotoxic and mutagenic to mammalian cells. Thus, a novel hypothesis to be tested in the proposed studies is that estrogen alters cell cycle regulation, proliferation in and damage to lens cells exposed to HZE particles (1 GeV/amu iron ion beams). The first Specific Aim shall be to define the dose-time interactions of high LET radiation (vs. low LET radiation) and estradiol to better understand the mechanism of estrogen action. The RBE for cataractogenesis in estrogen-treated animals exposed to1 GeV iron beams shall be determined using 60-Co gamma rays as a standard. The second specific aim shall be to determine the mechanism of estrogen-modulation of radiation-induced cataracts. Slit-lamp biomicroscopy will be used to measure lens opacification in exposed animals. Successful completion of these studies will lead to a greater understanding of the modulation of space radiation effects by hormones and novel intervention strategies for long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results obtained from this project may provide greater insight into the potential effects of hormones on normal tissues after radiotherapy treatments with high LET particles.
Task Progress:	This report summarizes progress during the second no-cost extension period for NASA Grant NNJ05HE64G. Work was focused on accruing data from experiments to complete the first Specific Aim of the project [to determine whether estrogen (17-beta-estradiol, E2) alters cataractogenesis induced by high LET radiation]. As described previously, we used our specially designed animal holder and collimator set-up to irradiate the right eyes of up to 3 rats simultaneously, while sparing the contralateral eye and most other normal tissues. This resulted in other structures receiving a dose of <3.5% of the dose to the target eye; no radiation-induced cataracts have ever been observed in the contralateral eye. The validity of our use of our animal holder and collimator set-up for small volume irradiations was established in previous Task Reports. During the final no-cost extension period, we finished accumulating data for experiments initiated as early as the Fall 2006 run at NSRL, at which time a radiation dose response experiment was started to determine whether modulation of cataractogenesis by E2 after exposure to high LET radiation is dependent upon the radiation dose. Ovariectomized or intact female 56 day old rats, or ovariectomized rats that received estrogen-implants were irradiated with 0.05 to 2 Gy of 600 MeV/amu 56Fe ions. Data accrual for the 1-2 Gy groups irradiated much earlier in project period was reported in previous Task Reports, but data accrual for animals irradiated with lower doses (0.5 Gy and less) required following the animals for their whole lifespan. The rats had also been irradiated midway through the project period due to scheduling problems. These latter issues chiefly contributed to the request for no-cost extensions. In essence, our recently accumulated data confirm that the presence or absence of hormones may modulate cataractogenesis induced by high LET radiation (600 MeV 56Fe ions). Furthermore, hormonal modulation of cataractogenesis is dependent upon radiation dose. Data for rats irradiated with doses of 600 MeV 56Fe ions ranging from 0.05 Gy to 2 Gy were accumulated. When anterior subcapsular (ASC) or posterior subcapsular (PSC) cataracts were scored for incidence and progression after irradiation, the influence of estrogen was best illustrated by comparing ovariectomized rats with rats that received E2 after ovariectomy, and intact rats with ovariectomized rats (bottom panel). A differential effect of endogenous and exogenous estrogen was observed over the range of doses when ovariectomized and E2-supplemented rats were compared, while the incidence rates of intact and ovariectomized were similar. However, ovariectomized rats with implanted E2 showed an increased rate of cataract progression, particularly at low doses (0.5 Gy and less), compared to overiectomized and intact rats exposed to endogenous estrogen. The trend for ASC cataracts was similar to that of PSC cataracts. Thus, the magnitude of the "estrogen effect" increased with decreasing dose. Thus, we conclude that endogenous or exogenous estrogen may enhance both the rate and incidence of cataractogenesis after exposure to a range of doses of high LET radiation. In fact, the presence of exogenous E2 appears to significantly modulate cataractogenesis induced by 56Fe ions in 56-day old rats, particularly after exposure to low doses. E2 reduced the time to 50% incidence of cataracts after exposure to iron ions, compared to ovariectomized control rats. Thus, E2 alters the dose-response relationship for cataractogenesis, and particularly enhances cataractogenesis induced by low doses of high LET radiation. The lowest doses used in our experiments (0.25 Gy and lower) are in the realm of doses relevant to astronauts involved in future space missions. This, along with knowledge of how E2 alters the latent period for radiation cataractogenesis, may aid in the assessment of risk estimates for cataractogenesis and other degenerative effects of space radiation.
Bibliography Type:	Description: (Last Updated: 07/17/2025)

Articles in Peer-reviewed Journals	Dynlacht JR, Valluri S, Garrett J, Mendonca MS, Lopez JT, Caperell-Grant A, Bigsby RM. "Age and hormonal status as determinants of cataractogenesis induced by ionizing radiation. I. Densely ionizing (high-LET) radiation." Radiat Res. 2011 Jan;175(1):37-43. Epub 2010 Nov 10. PubMed PMID: 21175345 , Jan-2011
Articles in Peer-reviewed Journals	Garrett J, Valluri S, Mendonca MS, Bigsby RM, Lopez J, Caperell-Grant A, Nees J, Dynlacht JR. "The protective effect of estrogen against radiation cataractogenesis is dependent upon the type of radiation." Radiat Res. 2020 Nov 10;194(5):557-65. https://doi.org/10.1667/RADE-20-00015.1 ; PMID: 33045089 , Nov-2020